

An Independent Angular-Statistics Test for Crystallographically Forbidden Symmetry in the Darb-i Imam Girih Tiling

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Abstract

Lu and Steinhardt (2007) showed that the girih tiling on the Darb-i Imam shrine (Isfahan, 1453 C.E.) is a nearly perfect quasicrystalline pattern, with crystallographically forbidden ten-fold orientational order, by mapping the tiling onto Penrose tiles and demonstrating quasi-periodic translational order. We ask a narrower, independent question by a completely different route. If the tile arrangement carries forbidden orientational symmetry, that symmetry should be visible in the angular statistics of the tile centroids alone, with no tile-to-Penrose mapping and no appeal to translational order. We extract tile centroids from an image, treat them as a point cloud, and measure the strength of n -fold rotational symmetry in the nearest-neighbour angular distribution. The discriminator is a single scalar, the forbidden excess, defined as the stronger of the five- and ten-fold signals minus the stronger of the four- and six-fold signals. It is positive for quasicrystalline arrangement and negative for periodic arrangement. The measurement is performed with an existing open-source instrument, SCOTT (Structural and Coherent Order in Topological Transforms), used here without domain-specific tuning. Against a calibrated set of synthetic controls, a Penrose tiling reads +1.14, two genuinely periodic tilings read -7.81 (square, four-fold) and -3.77 (hexagonal, six-fold), and white noise gives a trivial null. A girih reconstruction of the Darb-i Imam small-scale pattern reads +1.29 (full image at native aspect ratio, 406 motifs), on the quasicrystalline side with Penrose and far from both periodic controls. The result is an independent corroboration, by orientational statistics, of the forbidden symmetry Lu and Steinhardt identified. It is a screening test, not a constructive proof, and it is applied here to the authors' reconstruction as an instrument check. A test on a photograph of the shrine wall itself is the natural next step and is not attempted here.

1 Introduction

The conventional account of girih patterns in medieval Islamic architecture held that they were drafted as networks of zigzagging lines with straightedge and compass. Lu and Steinhardt [1] showed that by 1200 C.E. these patterns were being constructed instead as tessellations of a small set of decorated polygons, the girih tiles, and that by the fifteenth century the approach was combined with self-similar subdivision to produce nearly perfect quasi-crystalline tilings. Their central example is the Darb-i Imam shrine in Isfahan (1453 C.E.). They established its quasicrystalline character by mapping the girih tiling onto Penrose kites and darts, showing the mapping is consistent across the pattern with only a handful of point defects, and arguing that the ratio of tile frequencies approaches the golden ratio, the signature of quasi-periodic translational order.

A quasicrystal has two linked properties, quasi-periodic translational order and crystallographically forbidden orientational symmetry. Periodic crystals are limited to two, three, four, and six-fold rotational symmetry. Five-fold and ten-fold are forbidden in a periodic tiling, and their presence in a long-range ordered pattern is the hallmark of a quasicrystal. Lu and Steinhardt’s argument runs primarily through translational order, the Penrose mapping and the golden ratio of tile frequencies.

We test the other property, the forbidden orientational symmetry, by an independent route that touches neither the Penrose mapping nor the translational argument. If the tile arrangement carries ten-fold orientational order, that order is a property of where the tiles sit relative to one another. It should therefore be measurable from the tile centroids alone, as an angular statistic, with no knowledge of the tile shapes, the decoration, the Penrose correspondence, or the subdivision rule. This is a deliberately weaker and more local probe than the original analysis. It cannot establish quasi-periodicity, and it does not try to. What it can do is ask whether the forbidden symmetry is present in the raw arrangement, by a method with no overlap with the original, so that agreement constitutes genuine corroboration rather than a re-derivation.

The measurement is carried out with SCOTT (Structural and Coherent Order in Topological Transforms), an open-source image-analysis instrument applied in companion work to optical bokeh [2], natural and synthetic images under amplitude-adjusted surrogate testing [3], and real and synthetic faces [4]. The instrument is the tool, not the subject. The contribution of this note is the independent angular-statistics route and its calibration, not the software that runs it.

2 Method

2.1 From image to point cloud

The input is a greyscale image. Because the discriminator measures exact angles between tile centroids, the normalisation must preserve those angles. We therefore pad each image to a centred square with a constant fill equal to the image median, then downscale isotropically to a fixed 512×512 grid. The fixed grid makes motif counts and angular statistics comparable across inputs regardless of original size, and the pad-then-downscale order ensures the resize is isotropic, so a genuine n -fold angle in the source is not deformed. The median fill avoids injecting an artificial dark border that the extraction step could mistake for structure. The synthetic controls are generated square, so for them the padding is a no-op; it matters for the non-square reconstruction image, where an anisotropic stretch would otherwise bend the very angles being measured. The same path is applied to every specimen and control in this study.

Tile centroids are extracted by a deliberately simple pipeline. An Otsu threshold separates bright tile interiors from dark borders. A single iteration of 3×3 elliptical erosion separates touching tiles. Connected components are labelled, and a centroid is taken for each component whose area falls in the range 60 to 7000 pixels, with a relaxed fallback range applied if fewer than fifteen motifs survive the first pass. The result is a set of two-dimensional points, one per tile, carrying no information about tile shape, colour, or decoration. The analysis below sees only these points.

2.2 Angular symmetry of the point cloud

For each point we compute the angles to its six nearest neighbours, pooled across the whole cloud into a histogram of 72 bins of five degrees each. Five degrees divides evenly into the 36° spacing of ten-fold symmetry and the 72° of five-fold, so the expected peaks fall on exact bin centres rather than straddling boundaries. A pattern with genuine n -fold orientational symmetry concentrates angular mass at multiples of $360/n$ degrees. We measure the strength of n -fold symmetry as the

mean histogram density in the n exact bins at those expected angles, expressed as an excess over the uniform baseline,

$$S_n = \max\left(0, \frac{\bar{p}_{\text{peak}} - 1/B}{1/B}\right), \quad (1)$$

where $\bar{p}_{\text{peak}}$ is the mean normalised density in the n target bins and $B = 72$ is the bin count. The angle to a neighbour at row-column offset $(\Delta r, \Delta c)$ is taken as $\text{atan2}(\Delta c, \Delta r)$. The measurement uses the single exact bin at each expected angle, with no tolerance window. It is local, computed per point and pooled, so it is insensitive to the global image boundary in a way a whole-image Fourier method is not.

2.3 The discriminator

The quantity that separates quasicrystalline from periodic arrangement is the forbidden excess,

$$E = \max(S_5, S_{10}) - \max(S_4, S_6). \quad (2)$$

The forbidden orders five and ten are the quasicrystalline signature. The allowed orders four and six are the periodic signature. A positive E means the forbidden signal dominates, the arrangement is quasicrystal-like. A negative E means an allowed periodicity dominates, the arrangement is an ordinary crystal. A minimum of fifteen motifs is required for any reading, and angular statistics degrade below roughly thirty motifs, so motif count is reported alongside every result. The excess is a simple linear difference of measured angular signals, with no fitting, no training, and no free parameters, which keeps the discriminator interpretable and its behaviour traceable to the geometry it measures.

3 Controls

The discriminator is only meaningful if it reads strongly positive on a known quasicrystal, strongly negative on known periodic tilings, and null on structureless input. We locked the acceptance criteria before running the controls. A positive control must read $S_{10} \geq 1.5$ and $E > 0$. Each periodic negative control must produce at least 100 motifs and read $E \leq 0$. All controls are synthetic, generated from known geometry, so the ground truth is not in question. All run through the identical pipeline described above.

3.1 Positive control, Penrose

A Penrose rhombus tiling, generated by de Bruijn pentagrid dualisation, is the canonical two-dimensional quasicrystal with exact five-fold symmetry. It reads $S_{10} = 2.16$, $S_5 = 2.14$, and $E = +1.14$, clearing the locked gate. This reproduces across two independent sessions on different machine builds to three decimal places.

3.2 Negative controls, two periodic tilings

A genuinely periodic tiling cannot carry five- or ten-fold order, so it must read $E \leq 0$. We use two independent periodic geometries to test both crystallographically allowed symmetries, not just one. The first is a square lattice of flat tiles, which carries four-fold symmetry. It reads $S_4 = 8.78$ dominating, with $E = -7.81$. The second is a hexagonal lattice, which carries six-fold symmetry. It reads $S_6 = 4.22$ elevated, with $E = -3.77$. Both clear the locked acceptance. The instrument does

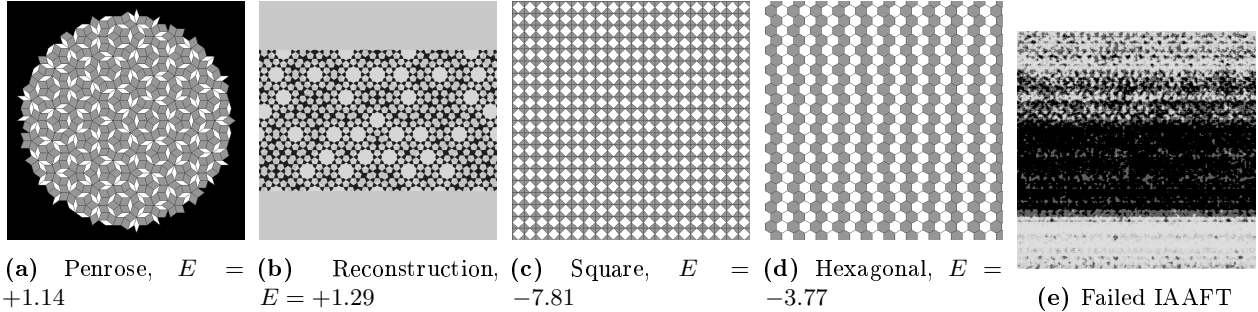


Figure 1: Inputs as the pipeline sees them, in greyscale at 512×512 after padding to square. All four working specimens (a-d) are flat two-tone tilings rendered in the same style so that extraction behaviour is matched. The two quasicrystalline inputs (a, b) and the two periodic controls (c, d) extract cleanly. The IAAFT surrogate (e) returns speckle that cannot be extracted, which is why a matched periodic tiling, not a surrogate, is the appropriate negative control here.

not merely register "periodic," it identifies which allowed symmetry is present, the square control reading four-fold dominant and the hexagonal control reading six-fold strong.

Both periodic controls are rendered in the same flat two-tone style as the Penrose control and the reconstruction specimen, so every input is the same visual class and extracts in the same way. This matters for the following reason.

3.3 A note on the surrogate that did not work

The natural first choice for a negative control was an iterative amplitude- adjusted Fourier transform (IAAFT) surrogate of the specimen, a randomisation that preserves the power spectrum and the intensity distribution while destroying spatial phase structure, used successfully on continuous-tone images in companion work [3]. On a flat-colour tiling it failed, and the failure is instructive. A flat tiling has a near-discrete intensity histogram, so the surrogate returns speckle that the Otsu extraction cannot resolve into tiles. The motif count collapsed from several hundred to twenty-nine, and the power spectrum was not in fact preserved, so the surrogate was not a valid IAAFT surrogate in the first place. We report this because it explains why the periodic controls above, matched to the specimen's visual class and extraction behaviour, are the appropriate negative controls for this class of image, and because the failed surrogate is part of the honest record. Figure 1 shows a failed surrogate alongside the working inputs.

4 Result on the Darb-i Imam reconstruction

We ran the test on a girih reconstruction of the small-scale Darb-i Imam pattern, the tile pattern of Lu and Steinhardt's Figure S7A rendered in flat colour. The reconstruction is the authors' own drawing, so this is explicitly an instrument check, a test of whether the method reads a known quasicrystalline arrangement correctly, not independent evidence about the shrine. The full image, padded to square and normalised to 512×512 , yields 406 extracted motifs and reads $S_{10} = 1.71$, $S_5 = 1.73$, $E = +1.29$. The ten-fold signal clears the gate, and the forbidden excess lands on the quasicrystalline side, above the Penrose value and far from both periodic controls.

Table 1 gives the full picture. The reconstruction and the Penrose control sit together on the quasicrystalline side of zero. The two periodic controls sit far on the crystalline side, via different allowed symmetries. White noise extracts no motifs and returns the trivial null. Every populated

Table 1: Forbidden excess E separates three arrangement classes on one axis. All values from a single machine, identical pipeline. Positive E is quasicrystalline, negative is periodic.

Specimen	Role	Motifs	S_{10}	S_4	S_6	E
Penrose	positive (quasicrystal)	671	2.16	1.02	0.08	+1.14
Darb-i Imam reconstruction	instrument check	406	1.71	0.43	0.17	+1.29
Hexagonal tiling	negative (periodic, 6-fold)	263	1.21	4.98	4.22	-3.77
Square tiling	negative (periodic, 4-fold)	400	0.95	8.78	2.23	-7.81
White noise	null	0	—	—	—	0.00

input is a flat tiling that extracts into hundreds of motifs, so the arrangement of the tiles is the only variable that differs between them, and the forbidden excess separates them cleanly.

5 Discussion

5.1 What this corroborates, and what it does not

The measurement supports one specific part of the Lu and Steinhardt result, the presence of crystallographically forbidden orientational symmetry in the tile arrangement, by a route that shares nothing with the original argument. It uses no Penrose mapping, no tile-frequency ratio, no subdivision rule, and no appeal to translational order. It reads only the angular relationships between neighbouring tile centroids. That a method with no overlap reaches the same qualitative conclusion is the substance of the corroboration.

The method does not establish quasi-periodicity, which is the harder and more important half of the quasicrystal definition. Forbidden orientational symmetry is necessary but not sufficient for a quasicrystal. Forbidden orientational order across a finite patch is noteworthy in its own right, but it does not by itself establish aperiodic long-range order, which is the harder claim. The translational argument in the original paper remains the load-bearing one for that claim. This note adds an independent check on the orientational half, nothing more.

5.2 Strength is sensitive to framing and scale

The ten-fold reading on the reconstruction, 1.71, is below the synthetic Penrose value of 2.16, but the forbidden excess is slightly higher, 1.29 against 1.14, because the reconstruction’s four- and six-fold content is low, so the excess reflects the separation between forbidden and allowed signals rather than the absolute strength of the forbidden signal. Earlier framings of the same pattern, and the earlier anisotropic-resize pipeline, read lower on the excess, between 0.46 and 1.22, depending on how much of the pattern the crop captured and how the normalisation set the tile scale. We also tested whether a tighter sub-crop would sharpen the reading. It did not. A sub-crop is downscaled harder by the fixed 512×512 grid, dropping the tile spacing below the extraction sweet spot, the motif count fell to 192 and the ten-fold signal weakened, although the forbidden excess remained faintly positive, the correct sign. The full image at its native aspect ratio, padded to square, is the appropriate input, and is the value reported above. The dependence on framing and scale is a real limitation of a motif-extraction method and is stated plainly rather than tuned away.

5.3 The photograph is the real test

The strongest version of this test is not the authors’ reconstruction but a photograph of the shrine wall itself. A photograph carries perspective skew, uneven lighting, surface wear, and decoration laid

over the tile boundaries, any of which can defeat the extraction step. We have not attempted it here. When we do, the pre-registered protocol is to set all preprocessing parameters on the reconstruction, where the answer is known, then apply them unchanged to the photograph, to run the identical pipeline on a periodic control which must stay negative, and to disclose every preprocessing step with before and after images. If the photograph reads positive under that protocol, it would be independent evidence about the shrine. If it does not, that is an acceptable and reportable limit of a motif-extraction method on worn architectural photography, and the corroboration would rest on the reconstruction and the matched controls. We state this in advance so that a positive photograph result cannot later be assembled by tuning.

5.4 Pre-registration

The acceptance criteria for the controls and the direction of the reconstruction prediction were locked before the runs. The positive gate ($S_{10} \geq 1.5$, $E > 0$) and the negative-control acceptance (at least 100 motifs, $E \leq 0$) were fixed in advance. During the build of the square control, the first run produced only 64 motifs and failed the motif floor, while already reading a strongly negative excess. We did not adjust the geometry to rescue the count, since the negative excess was fixed by the lattice and could not change sign on the arrangement axis. We set the tile density by arithmetic, choosing the cell count to land in the reconstruction’s own documented motif band, and recorded the second run as it came. This and the failed IAAFT surrogate are part of the record rather than edited out of it.

6 Reproducibility

The instrument, the control generators, and the run script are open source. The two periodic control generators and the script that runs the full calibration were added to the repository for this work. The calibration in Table 1 is regenerated from a single command, `python scripts/run_phase5b_calibration.py`, and the per-run output, including motif counts and all fold signals, is written to a JSON record in the project. The Penrose calibration reproduced across two sessions on different builds to three decimal places, and the control calibration reproduced on an independent machine the same day it was generated. The Darb-i Imam reconstruction image used here is the flat-colour rendering of the small-scale S7A pattern, and the source image dimensions and crop are recorded with the result.

7 AI use disclosure

AI language models were used throughout this work. Claude (Anthropic) was used for code development, for running and checking the control calibrations, for statistical and methodological review, and for drafting this note. GPT-4o Vision (OpenAI) has been used elsewhere in the SCOTT project as a crop-selection tool. The experimental design, the locked predictions, the choice of controls, the interpretation of every result, and the decision to report the failed IAAFT surrogate and the scale limit rather than omit them are the author’s. The author is not a mathematician or physicist by training, but has thirty years of experience building and managing software systems, and works with AI tools as a routine part of that practice. The author’s role was to direct the investigation, make the judgment calls, and take responsibility for the claims. A fuller account of AI contributions across the SCOTT project is available in the repository at <https://github.com/scottcundill/scott-framework>.

8 Data and code availability

The SCOTT (Structural and Coherent Order in Topological Transforms) source code, the control generators, the calibration run script, and the JSON records of all runs reported here are available in the project repository at <https://github.com/scottcundill/scott-framework>. The Darb-i Imam reconstruction derives from the Supporting Online Material of Lu and Steinhardt [1].

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